

5. (20 points) There are three types of coins A, B, C : coin A tosses head with probability p_1 , coin B tosses head with probability p_2 , coin C tosses head with probability p_3 . The experimenter selects one of the three coins at random (with probability $1/3$ each), and then tosses it independently 6 times. What is the probability that the experimenter see 4 many heads and 2 many tails?

Say X is the chosen point.

$$P(4H \ 2T) = P(4H \ 2T \cap \{X=A\}) + P(4H \ 2T \cap \{X=B\}) + P(4H \ 2T \cap \{X=C\}) =$$

$$\begin{aligned} &\rightarrow \underbrace{P(4H \ 2T | X=A)}_{\substack{\downarrow \\ \text{Binomial distribution} \\ \text{with } p_1}} \cdot \underbrace{P(X=A)}_{\frac{1}{3}} + \underbrace{P(4H \ 2T | X=B)}_{\substack{\downarrow \\ \left(\binom{6}{2} p_2^4 (1-p_2)^2\right)}} \cdot \underbrace{P(X=B)}_{\frac{1}{3}} + \underbrace{P(4H \ 2T | X=C)}_{\substack{\downarrow \\ \left(\binom{6}{2} p_3^4 (1-p_3)^2\right)}} \cdot \underbrace{P(X=C)}_{\frac{1}{3}}. \end{aligned}$$

$$= \frac{1}{3} \left(15 p_1^4 (1-p_1)^2 + 15 p_2^4 (1-p_2)^2 + 15 p_3^4 (1-p_3)^2 \right)$$

Ex (old quiz problem) X is a discrete R.V. with mgf.

$$M(t) = \frac{3e^t}{4} \left(1 - \frac{e^t}{4} \right)^{-1}$$

(a) EX A) 1 B) $\frac{4}{3}$ C) $\frac{3}{4}$ D) $\frac{8}{3}$ E) $\frac{1}{2}$

(b) $\text{Var } X$ A) 1 B) $\frac{2}{3}$ C) $\frac{3}{2}$ D) $\frac{4}{9}$ E) $\frac{9}{4}$

(c) Support and pmf of X .

Solution: Remember

$$EX = M'(0)$$

$$EX^2 = M''(0)$$

$$\begin{aligned} \text{Var}(X) &= EX^2 - (EX)^2 \\ &= M''(0) - (M'(0))^2 \end{aligned}$$

$$M(t) = \frac{\frac{3}{4}e^t}{1 - \frac{e^t}{4}} = \frac{A}{B} \quad M'(t) = \frac{A'B - AB'}{B^2}$$

$$\begin{aligned} M'(t) &= \frac{\frac{3}{4}e^t \cdot (1 - \frac{e^t}{4}) - \frac{3}{4}e^t(-\frac{1}{4}e^t)}{(1 - \frac{e^t}{4})^2} \\ &= \frac{\frac{3}{4}e^t \left(1 - \frac{e^t}{4} + \frac{1}{4}e^t\right)}{(1 - \frac{e^t}{4})^2} \end{aligned}$$

$$= \frac{3}{4}e^t \cdot (1 - \frac{e^t}{4})^{-2}$$

$$EX = M'(0) = \frac{3}{4} \cdot 1 \cdot (1 - \frac{1}{4})^{-2} = \frac{3}{4} \cdot \left(\frac{3}{4}\right)^{-2} = \frac{3}{4} \cdot \frac{16}{9} = \boxed{\frac{4}{3}}$$

$$\text{Var}(X) = M''(0) - \underbrace{(M'(0))^2}_{\left(\frac{4}{3}\right)^2} = M''(0) - \frac{16}{9}$$

$$M'(t) = \frac{3}{4}e^t (1 - e^t)^{-2}$$

$$M'(t) = \underbrace{\frac{3}{4} e^t}_A \underbrace{\left(1 - \frac{e^t}{4}\right)^{-2}}_B$$

$$M''(t) = \underbrace{\frac{3}{4} e^t}_{A'} \underbrace{\left(1 - \frac{e^t}{4}\right)^{-2}}_B + \underbrace{\frac{3}{4} e^t}_A \underbrace{(-2) \left(1 - \frac{e^t}{4}\right)^{-3} \cdot \frac{-e^t}{4}}_{B'}$$

$$M''(0) = \frac{3}{4} \cdot \left(1 - \frac{1}{4}\right)^{-2} + \frac{3}{4} \cdot 1 \cdot (-2) \left(1 - \frac{1}{4}\right)^{-3} \cdot \frac{-1}{4}$$

$$= \frac{3}{4} \cdot \left(\frac{3}{4}\right)^{-2} + \frac{3}{4} \cdot \frac{1}{2} \cdot \left(\frac{3}{4}\right)^{-3}$$

$$= \frac{4}{3} + \frac{1}{2} \left(\frac{4}{3}\right)^2 = \frac{4}{3} + \frac{1}{2} \cdot \frac{16}{9} = \frac{20}{9}$$

$$\text{Var}(X) = M''(0) - (M'(0))^2 = \frac{20}{9} - \left(\frac{4}{3}\right)^2 = \frac{20-16}{9} = \boxed{\frac{4}{9}}$$

(c) formula $1 + r + r^2 + \dots = \frac{1}{1-r} \leftarrow$

$$M(t) = \frac{3}{4} e^t \cdot \underbrace{\frac{1}{1 - \frac{1}{4} e^t}}_{\substack{\frac{1}{1-r} \\ r = \frac{1}{4} e^t}}$$

$$= \frac{3}{4} e^t (1 + r + r^2 + \dots)$$

$$= \frac{3}{4} e^t \left(1 + \frac{e^t}{4} + \frac{e^{2t}}{4^2} + \frac{e^{3t}}{4^3} + \dots\right)$$

$$= \frac{3}{4} e^t + \frac{3}{4} \frac{e^{2t}}{4} + \frac{3}{4} \frac{e^{3t}}{4^2} + \dots$$

$$= \frac{3}{4} e^t + \frac{3}{4^2} e^{2t} + \frac{3}{4^3} e^{3t} + \dots$$

$$= \sum_{x \geq 1} \frac{3}{4^x} e^{tx} \Rightarrow S_X = \{1, 2, 3, \dots\}$$

Remember dice example.

$$M_{pf_D}(t) = \frac{1}{6} e^t + \frac{1}{6} e^{2t} + \dots + \frac{1}{6} e^{6t}$$

$$S_D = \{1, 2, 3, \dots, 6\}$$

$$f(x) = \frac{1}{6}$$

$$= \sum_{x \geq 1} \frac{3}{4^x} e^{tx} \Rightarrow S_X = \{1, 2, 3, \dots\}$$

$$\downarrow$$

$$f(x) = \frac{3}{4^x}$$

$$f(1) = \frac{3}{4} \quad f(2) = \frac{3}{4^2} \quad f(3) = \frac{3}{4^3} \dots$$

You have a coin that comes heads with $p = \frac{3}{4}$

X is the first throwing we see a head.

$$P(X=1) = P(H) = \frac{3}{4}$$

$$P(X=2) = P(TH) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{4^2}$$

$$P(X=3) = P(TTH) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{4^3}$$

$$P(X=3) = P(TTH) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{4^3}$$

$$P(X=n) = P(\underbrace{TT \dots T}_{n-1} H) = \underbrace{\frac{1}{4} \cdot \frac{1}{4} \cdot \dots \frac{1}{4}}_{n-1} \cdot \frac{3}{4} = \frac{3}{4^n}$$

→ Realize this has the same pmf as the example above

$$X \sim \text{Geo}\left(\frac{3}{4}\right)$$

For Geometric random variables with parameter p , $X \sim \text{Geo}(p)$.

$$EX = \frac{1}{p} \quad \left. \vphantom{EX} \right\} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$$

$$\text{Var } X = \frac{1-p}{p^2} \quad \left. \vphantom{\text{Var } X} \right\} = \frac{1-\frac{3}{4}}{\left(\frac{3}{4}\right)^2} = \frac{4}{9}$$

$$\text{Mgf}(X) = \frac{p e^t}{1 - (1-p)e^t}$$

$$M(t) = \frac{\frac{3}{4} e^t}{1 - \frac{1}{4} e^t} = \leftarrow \text{this form when } p = \frac{3}{4}.$$

X is Geometric RV with $p = \frac{3}{4}$.

Geometric RV

Have a coin with probability coming heads p .

X = The first time we have a head.

$$S_X = \{1, 2, 3, \dots\}$$

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$$f(x) = P(X=x) = P(\underbrace{T T T \dots T}_{x-1} H) = \underbrace{\left(\frac{1}{9}\right)^{x-1}}_p \cdot p$$

$$= 9^{x-2} \cdot p.$$

$$p = \frac{1}{9} \quad EX = \frac{1}{p} = 9$$

Example 1.18. (Old Quiz Problem-PSI 2.6.10-modified) Red Rose Tea randomly began placing one of 25 English porcelain miniature figurine in a 100-bag box of tea, ~~selecting from ten figurine in the American Heritage series.~~ A customer, a big fan of George Washington, wants to have three copies of George Washington figurine; one for his collection, one to give to an acquaintance, and one for his own entertainment. On the average, how many boxes of tea must be purchased by a customer to obtain three copies of George Washington figurine? are

A) 10 B) 25 C) 50 D) 625 E) 100

Sol: X is the number of tea boxes required to get GWF.

$$P(X=1) = \frac{1}{25}$$

$$P(X=2) = \frac{24}{25} \cdot \frac{1}{25}$$

$$P(X=3) = \frac{24}{25} \cdot \frac{24}{25} \cdot \frac{1}{25} = \left(\frac{24}{25}\right)^2 \cdot \frac{1}{25}$$

$$X \sim \text{Geo}\left(\frac{1}{25}\right) \quad EX = \frac{1}{\frac{1}{25}} = 25$$

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Solution: Now, this problem is actually negative binomial problem.

$X = \# \text{ of tea boxes to get three GWF}$

$X \sim \text{Neg Bin}(p = \frac{1}{25}, r = 3)$

$$E[X] = \frac{r}{p} = \frac{3}{\frac{1}{25}} = 75$$



Negative binomial is actually sum of geometric RVs.